open vs code:

create new repository (or) choose clone repository

ctrl + shift + P

Paste link

git linked with VScode

open Terminal → new terminal →

git init #initializing git

git add filename # for changes in prog

git commit -m "ver 1" # to find version changes

git push → go to git repo

changes updated

1) calculator program

Interminal

git init

git add filename

---

notepad kavitha.txt

notepad " .py

git add (space) .

git status

git commit -m "adding the python programs"

New terminal

git clone link

cd team

git checkout br3

git add .

git commit -m "any name"

git push origin br3

git checkout main

git merge br3

git push origin main

CI / CD Pipeline

Using Jenkins & Maven

maven.apache.org/download.cgi

Binary zip archive → (2nd) → bin.zip

copy (folder)

paste in jdk folder

extract (delete after goal)

after extracted

copy address

Edit Environment variable

System variable → path (Edit) → New

(paste the path) ok

New

Variable name: maven_home (caps)

: paste path

Ok → New

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github

cmd prompt

setting username & mailid in system:

```
git config - - global user.name "Nila-kau"
git config - - global user.email "[illegible]"
```

VS code

Open the folder for Git

↓

New Terminal

↓

```
git clone "repo link"
cd reponame
```

to move for staging area

```
git status ⇒ untracked files present
                (∵ created in VS code manually)
git add newfilename ⇒ Adding to git from VS code
git commit -m "Test 2.txt file added"
git push origin main
```

from VS code to git

```
git pull
```

to update changes from git to VS code

Github ⇒ Insert/any modification ⇒ Add ⇒ commit ⇒ push

new folder in vs code under gitfolder

cd .. // coming back
cd filefoldersname
git init
git add . // take all modified files to staging area
git status
git commit -m "created index.html and style.css".
git remote add origin repolink
git branch ⇒ *master
Error ⇒ git push origin main
error ⇒ git push origin master (or)
git branch -M main
git branch

All codes in vs code pushed ⇐ git push origin main into git repo.

cmd: clear

Branching

In git
main
↓ (dropdown)
create new branch

say new branch name gitbranch1

In VS terminal,

```
git pull
git branch -a // to view all branches (with current branch highlighted)
git checkout gitbranch1 // switched to new branch gitbranch1
git diff gitbranch1
git merge gitbranch1
git push origin main
```

git restore filename

git remote set-url origin repolink
git branch -M main
git push -u origin main

Pull request (git feature)

clear

```
git branch newbranch2
git branch -a // view all branches with current working branch highlighted (main)
git checkout newbranch2 // switched to this branch
git branch
```

// create a file under the repo and name newbranch2.txt then give

```
git add.
git commit -m "Added newbranch2 text file"
git push origin newbranch2
```

cmd: clear

Branching

In git
main
↓ (dropdown)
create new branch

say new branch name gitbranch1

In VS terminal,

```
git pull
git branch -a // to view all branches (with current branch highlighted)
git checkout gitbranch1 // switched to new branch gitbranch1
git diff gitbranch1
git merge gitbranch1
git push origin main
```

git restore filename

git remote set-url origin repolink
git branch -M main
git push -u origin main

Pull request (git feature)

clear

```
git branch newbranch2
git branch -a // view all branches with current working branch highlighted (main)
git checkout newbranch2 // switched to this branch
git branch
```

// create a file under the repo and name newbranch2.txt then give

```
git add.
git commit -m "Added newbranch2 text file"
git push origin newbranch2
```

Jenkins

① create freestyle job

New Item → name: Hello-jenkins (or) kavitha

↓

Free style project ⇒ ok

configure → Environment → command

↓

echo Hello Jenkins

Apply → save

console output

build history

② create github repository

- repo name: jenkins-demo

visibility → public

Readme → off

③ configure git in jenkins

Jenkins → manage jenkins → Tools

path to Git executable

c: \Program Files \Git \bin \git

↓

Apply ⇒ Save

④ Jenkins → New Item : git-Job (Freestyle project)

↓

ok

Git → Job ⇒ configure ⇒ source code management

↓

Paste git repo url

↓

Apply save

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⑤ Git - Job ⇒ configure ⇒ Environment ⇒
command ⇒ echo Echo pulled Succ
(In build step)
dir
↓
Apply ⇒ Save

⑥ Triggers ⇒ Pool SCM (tick) ⇒ # /5 * * * *
(or) web Hook triggers

⑦ Build automatically on code change ⇒ status

① Create multiple Freestyle jobs:
Job-A (Build)
Jenkins → All → New Item → Job A (item name)
↓
Freestyle project (ok)

Job-B (test)
Job-C (deploy)

② Use
Job A → configure → Environment →
Build steps → command
echo Building the project
↓
Apply → Save

Job B ⇒ echo Running tests
Job C ⇒ echo Deploying application

③ Job chaining
Jenkins ⇒ Job A ⇒ configure → Build steps ⇒
post build action → Job B
Job B ⇒ " " " ⇒ Job C

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Install pipeline stage view:
Jenkins → Manage Jenkins → Plugins
⇓
Download progress

Observe stepwise execution
Job A → status
console output

Create pipeline job
new item : pipeline-Job
(pipeline) → ok

① configure → pipeline
⇓
Script →
save

Pipeline-Job
② Status
③ console output

Git → jenkins-demo / Jenkinsfile clone
Jenkins → New Item : Jenkinsfile-pipeline
(pipeline)
ok

configure : pipeline section
Repo Url
credentials : kaushik/****
Branch specifier : */main
save

Status.

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Install pipeline stage view:
Jenkins → Manage Jenkins → Plugins
⇓
Download progress

Observe stepwise execution
Job A → status
console output

Create pipeline job
new item : pipeline-Job
(pipeline) → ok

① configure → pipeline
⇓
Script →
save

Pipeline-Job
② Status
③ console output

Git → jenkins-demo / Jenkinsfile clone
Jenkins → New Item : Jenkinsfile-pipeline
(pipeline)
ok

configure : pipeline section
Repo Url
credentials : kaushik/****
Branch specifier : */main
save

Status.

Maven project.

new repo =>

pom.xml

src/main/java/com/example/App.java

src/test/java/com/example/AppTest.java

push & clone repo in vs code => git clone repolink

mvn compile => Build success

mvn test

mvn clean install

create & run jenkins pipline:

new item: MavenPipeline (pipeline) => ok

pipeline => def: Pipeline script

(paste code in script)

save

Status

—x—